RANA ABDULLAH AZHAR

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PROFESSIONAL SUMMARY

Electrical Engineer with strong focus on embedded and instrumentation systems, IoT architecture, and industrial automation. Hands-on experience in implementing smart city solutions using LoRaWAN, MQTT, Modbus TCP/IP, and Wi-Fi communication. Proven capability in sensor integration and calibration, embedded system integration, hardware debugging, and real-time monitoring systems. Solid academic and practical foundation in power electronics, control systems, RF and communication systems, including battery management systems, DC-DC converters, motor control, and antenna design.

EDUCATION

National University of Sciences and Technology (NUST) Islamabad, Pakistan
Bachelor of Engineering in Electrical Engineering *Nov 2021 – May 2025*

- Relevant Coursework: Power Electronics, Embedded Systems, Control Systems, DSP, Communication Systems, Electrical Machines

Army Public School and College Ordnance Road Rawalpindi, Pakistan
Intermediate (Pre-Engineering) *July 2019 – Sep 2021*

Army Public School and College Ordnance Road Rawalpindi, Pakistan
Matriculation (Science) *March 2017 – April 2019*

PROFESSIONAL EXPERIENCE

Smart TechOne Rawalpindi, Pakistan
Junior IoT Engineer *July 2025 – Aug 2026*

- Deployed and maintained 50+ IoT sensor nodes and LoRaWAN gateways for smart city applications, achieving 98% uptime
- Configured and managed LoRaWAN communication using MQTT-based data transmission
- Implemented Modbus TCP/IP communication for embedded water quality sensors measuring pH, EC, dissolved oxygen, chlorine, ORP, chlorophyll, and turbidity
- Troubleshot networking, embedded hardware, and communication issues across distributed real-time systems

PROJECTS

Smart City IoT Infrastructure Deployment (Professional Project) July 2025 – Aug 2026

- Deployed embedded IoT sensor networks for smart waste management, traffic monitoring, weather, and air quality systems
- Utilized LoRaWAN and MQTT protocols for reliable wireless communication
- Integrated embedded devices with real-time dashboards for monitoring and analytics
- Performed system-level debugging to ensure high availability and data accuracy

Smart Traffic Light and Surveillance Integration (Professional Project) Aug 2025 – Oct 2025

- Developed IoT workflows for smart traffic light systems using real-time sensor inputs
- Integrated multiple communication protocols into a unified monitoring platform

Water Quality Monitoring System (Professional Project) July 2025 – Aug 2025

- Designed an embedded monitoring system using Modbus TCP/IP for industrial-grade water quality sensors
- Integrated pH, EC, dissolved oxygen, chlorine, ORP, chlorophyll, and turbidity sensors
- Focused on reliable communication, sensor accuracy, and real-time data availability

Wireless Holter Monitor (Final Year Project) Feb 2024 – March 2025

- Developed an embedded wireless Holter monitor for continuous heart rate monitoring
- Implemented real-time data transmission for biomedical applications

Battery Management System (BMS) Feb 2024 – May 2024

- Designed an embedded Battery Management System for lithium-ion battery packs
- Implemented voltage, current, and temperature monitoring with protection logic

Audio Amplifier Circuit Design Nov 2023 – March 2024

- Designed and simulated a discrete analog audio amplifier using BJTs and MOSFETs
- Optimized gain, bandwidth, and frequency response for improved audio performance

Closed-Loop Buck Converter Design and Control Sept 2023 – Dec 2023

- Designed and implemented a closed-loop DC-DC buck converter
- Applied feedback-based voltage regulation and validated performance

DC Motor Position Control Using PID Sept 2023 – Nov 2023

- Implemented PID-based position control for a DC motor
- Analyzed system stability and transient response

Single-Phase Transformer Design Sept 2022 – Nov 2022

- Designed a single-phase transformer with multiple voltage taps
- Evaluated efficiency, regulation, and load performance

FM Transmitter Design Feb 2022 – April 2022

- Designed and implemented an FM transmitter circuit
- Studied modulation techniques and frequency stability

Microstrip Patch Antenna Design (2.4 GHz) Sept 2021 – Dec 2021

- Designed and simulated a microstrip inset-fed square patch antenna at 2.4 GHz
- Validated antenna performance using HFSS and fabricated PCB prototype

TECHNICAL SKILLS

IoT & Industrial Communication: LoRaWAN, Modbus, Wi-Fi Communication

Embedded & Instrumentation Systems: Embedded System Integration, Sensor Interfacing and Calibration, Hardware Debugging, Real-Time Monitoring Systems

Control Systems: PID Control (P/PI/PD/PID), Closed-Loop Control Systems, DC Motor Control, Feedback System Analysis

Power & Electronics: Power Electronics, DC-DC Buck Converters, Analog Circuit Design, Power Supply Design, Single-Phase Transformer Design

RF & Communication Systems: FM Transmitter Design, Antenna Design (Microstrip Patch), RF Fundamentals, Communication Systems

Engineering Tools & Simulation: MATLAB/Simulink, Proteus, NI Multisim, HFSS, Arduino, MS Office

CERTIFICATIONS

- Junior IoT Engineer Certificate — Smart TechOne July 2025 – Aug 2026